

Supplementary Material: Traits predict forest phenological responses to photoperiod more than temperature

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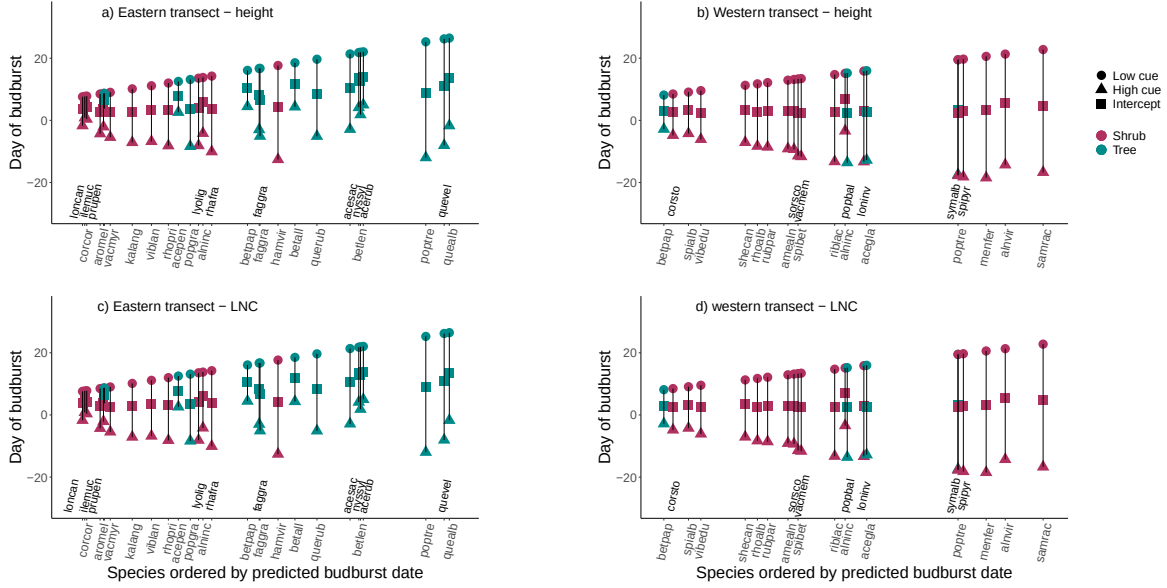


Figure S1: We found the estimated day of budburst differed between our full model (intercept plus cues, depicted as triangles for high cues and as circles for low cues), versus the intercepts only model (without cues, shown as squares). Species are ordered in increasing budburst dates for both the (a, c) eastern (b, d) and western populations, and in general span from early budbursting shrubs shown in red, to late budbursting trees shown in blue. (a, b) For traits such as height we found distinct partitioning of budburst across shrub and tree species, but this was not the case for all traits, (c, d) with the ranked order of species in our model of leaf nitrogen content exhibiting a highly mixed budburst order for our shrub and tree species.

Table S1: List of species names and abbreviations, as well as the transect in which each species was sampled from and plant architecture.

Species	Spp. abbreviation	Transect	Plant architecture
<i>Acer glabrum</i>	acegla	Western	Tree
<i>Acer pensylvanicum</i>	acepen	Eastern	Tree
<i>Acer rubrum</i>	acerub	Eastern	Tree
<i>Acer saccharum</i>	acesac	Eastern	Tree
<i>Alnus incana</i>	alninc	Eastern and Western	Shrub
<i>Alnus viridis</i>	alnvir	Western	Shrub
<i>Amelanchier alnifolia</i>	amealn	Western	Shrub
<i>Aronia melanocarpa</i>	aromel	Eastern	Shrub
<i>Betula alleghaniensis</i>	betall	Eastern	Tree
<i>Betula lenta</i>	betlen	Eastern	Tree
<i>Betula papyrifera</i>	betpap	Eastern	Tree
<i>Betula papyrifera</i>	betpap	Eastern and Western	Tree
<i>Corylus cornuta</i>	corcor	Eastern	Shrub
<i>Cornus stolonifera</i>	corsto	Western	Shrub
<i>Fagus grandifolia</i>	faggra	Eastern	Tree
<i>Fraxinus nigra</i>	franig	Eastern	Tree
<i>Hamamelis virginiana</i>	hamvir	Eastern	Shrub
<i>Ilex mucronata</i>	ilemuc	Eastern	Shrub
<i>Kalmia angustifolia</i>	kalang	Eastern	Shrub
<i>Lonicera canadensis</i>	loncan	Eastern	Shrub
<i>Lonicera involucrata</i>	loninv	Western	Shrub
<i>Lyonia ligustrina</i>	lyolig	Eastern	Shrub
<i>Menziesia ferruginea</i>	menfer	Western	Shrub
<i>Nyssa sylvatica</i>	nyssyl	Eastern	Tree
<i>Populus balsamifera</i>	popbal	Western	Tree
<i>Populus grandidentata</i>	popgra	Eastern	Tree
<i>Populus tremuloides</i>	poptre	Eastern and Western	Tree
<i>Prunus pensylvanica</i>	prupen	Eastern	Tree
<i>Quercus alba</i>	quealb	Eastern	Tree
<i>Quercus rubra</i>	querub	Eastern	Tree
<i>Quercus velutina</i>	quevel	Eastern	Tree
<i>Rhamnus frangula</i>	rhafran	Eastern	Shrub
<i>Rhododendron albiflorum</i>	rhoalb	Western	Shrub
<i>Rhododendron prinophyllum</i>	rhopri	Eastern	Shrub
<i>Ribes lacustre</i>	riblac	Western	Shrub
<i>Rubus parviflorus</i>	rubpar	Western	Shrub
<i>Sambucus racemosa</i>	samrac	Western	Shrub
<i>Shepherdia canadensis</i>	shecan	Western	Shrub
<i>Sorbus scopulina</i>	sorsco	Western	Shrub
<i>Spiraea alba</i>	spialb	Western	Shrub
<i>Spiraea betulifolia</i>	spibet	Western	Shrub
<i>Spiraea pyramidata</i>	spipyr	Western	Shrub
<i>Symphoricarpos albus</i>	symalb	Western	Shrub
<i>Vaccinium membranaceum</i>	vacmem	Western	Shrub
<i>Vaccinium myrtilloides</i>	vacmyr	Eastern	Shrub
<i>Viburnum cassinoides</i>	vibcas	Eastern	Shrub
<i>Viburnum edule</i>	vibedu	Western	Shrub
<i>Viburnum lantanoides</i>	viblan	Eastern	Shrub

Table S2: We obtained the temperature data we used to calculate field chilling from the nearest weather station to each site. Temperature data for the fall of 2019 and winter 2020 was obtained from the Government of Canada’s Environmental and natural resources’ historical data, using the Hope Slide weather station for our E.C. Manning park samples and the Smithers airport weather station for our Smithers samples (Government of Canada: Environment and natural resources, 2026). For our eastern samples, weather data for the fall of 2014 and winter 2015 was obtained from weather stations at Harvard Forest (Boose et al., 2026) and in the Government of Canada’s Environmental and natural resources’ historical data from the St. Hippolyte weather station (Government of Canada: Environment and natural resources, 2026). Listed below are the total chill portions that include both field and experimental chilling. Chill portions were calculated using the Dynamic Model in the chillR package (v. 0.73.1, Luedeling, 2020).

Site	Chilling treatment	Chill portions
Harvard forest	Field chilling	56.62
Harvard forest	Field chilling + 30 d at 4°C	94.06
St. Hippolyte	Field chilling	44.63
St. Hippolyte	Field chilling + 30 d at 4°C	82.06
Smithers	Field chilling + 21 d at 4°C	54.95
Smithers	Field chilling + 56 d at 4°C	74.67
Manning park	Field chilling + 21 d at 4°C	55.09
Manning park	Field chilling + 56 d at 4°C	75.33

Table S3: Summary output from a joint Bayesian model of height and budburst phenology in which species are partially pooled. The model includes environmental cues as z -scored continuous variables, allowing comparisons to be made across cues.

	Mean	5%	25%	75%	95%
Transect	7.58	-1.08	4.05	11.08	16.16
Latitude	0.05	-0.03	0.02	0.08	0.12
Transect x latitude	-0.20	-0.39	-0.28	-0.12	-0.01
Forcing	-10.31	-12.59	-11.09	-9.41	-8.35
Chilling	-13.44	-17.22	-14.81	-11.99	-9.89
Photoperiod	-2.54	-4.03	-3.20	-1.94	-0.81
Trait x forcing	0.28	-0.07	0.14	0.42	0.61
Trait x chilling	0.25	-0.46	-0.04	0.54	0.96
Trait x photoperiod	-0.34	-0.52	-0.41	-0.27	-0.16

Table S4: Summary output from a joint Bayesian model of diameter and budburst phenology in which species are partially pooled. The model includes environmental cues as z -scored continuous variables, allowing comparisons to be made across cues

	Mean	5%	25%	75%	95%
Transect	-24.13	-42.56	-31.70	-16.66	-5.88
Latitude	0.16	0.00	0.09	0.22	0.31
Transect x latitude	0.52	0.11	0.35	0.68	0.92
Forcing	-9.14	-11.41	-10.07	-8.30	-6.53
Chilling	-12.52	-16.21	-14.03	-11.12	-8.55
Photoperiod	-3.79	-5.68	-4.46	-3.08	-2.01
Trait x forcing	0.20	-0.06	0.09	0.30	0.46
Trait x chilling	0.16	-0.34	-0.04	0.37	0.65
Trait x photoperiod	-0.20	-0.34	-0.26	-0.15	-0.07

Table S5: Summary output from a joint Bayesian model of branch wood density and budburst phenology in which species are partially pooled. For this model the trait data was rescaled by 10 and environmental cues were included as z -scored continuous variables, allowing comparisons to be made across cues.

	Mean	5%	25%	75%	95%
Transect	0.72	0.44	0.60	0.84	1.00
Latitude	0.004	0.001	0.002	0.005	0.006
Transect x latitude	-0.015	-0.021	-0.017	-0.012	-0.009
Forcing	-14.91	-22.87	-17.59	-11.74	-8.63
Chilling	-20.94	-33.15	-25.21	-16.26	-9.75
Photoperiod	-0.61	-4.12	-2.14	0.82	3.30
Trait x forcing	1.73	-0.26	0.89	2.53	3.75
Trait x chilling	2.66	-0.94	1.32	4.04	6.11
Trait x photoperiod	-0.93	-1.99	-1.38	-0.51	0.19

Table S6: Summary output from a joint Bayesian model of leaf mass area and budburst phenology in which species are partially pooled. For this model the trait data was rescaled by 100 and environmental cues were included as z -scored continuous variables, allowing comparisons to be made across cues.

	Mean	5%	25%	75%	95%
Transect	-0.22	-0.27	-0.24	-0.20	-0.18
Latitude	-0.0015	-0.0020	-0.0017	-0.0014	-0.0011
Transect x latitude	0.0048	0.0038	0.0044	0.0052	0.0058
Forcing	-2.68	-19.52	-9.07	4.01	13.94
Chilling	-2.67	-30.26	-13.87	8.06	26.04
Photoperiod	-13.97	-23.12	-18.10	-10.12	-3.54
Trait x forcing	-0.59	-1.97	-1.15	-0.04	0.83
Trait x chilling	-0.87	-3.29	-1.80	0.09	1.43
Trait x photoperiod	0.89	0.01	0.57	1.24	1.65

Table S7: Summary output from a joint Bayesian model of leaf nitrogen content and budburst phenology in which species are partially pooled. The model includes environmental cues as z -scored continuous variables, allowing comparisons to be made across cues.

	mean	5%	25%	75%	95%
Transect	-4.71	-12.82	-8.03	-1.40	3.24
Latitude	-0.14	-0.21	-0.17	-0.11	-0.06
Transect x latitude	0.04	-0.13	-0.03	0.11	0.22
Forcing	-14.43	-32.60	-21.41	-7.56	3.76
Chilling	-35.11	-68.11	-47.92	-22.15	-4.07
Photoperiod	6.39	-2.39	2.55	9.84	16.53
Trait x forcing	0.45	-1.26	-0.19	1.13	2.04
Trait x chilling	2.07	-0.81	0.96	3.24	4.82
Trait x photoperiod	-0.91	-1.69	-1.22	-0.61	-0.11

4 References

- 5 Boose, E., E. Gould, and M. VanScoy. 2026. Harvard Forest Data Archive: HF300 (v.14). Environ-
6 mental Data Initiative.
- 7 Government of Canada: Environment and natural resources. 2026. Historical Data.
- 8 Luedeling, E. 2020. chillR: Statistical Methods for Phenology Analysis in Temperate Fruit Trees.
9 <https://CRAN.R-project.org/package=chillR>.